Study Research Note

Navier-Stokes Equations & Numerical Methods

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1 Background

Recently, OpenAI announced an article about the Navier-Stokes equation [1]. To understand it, I had to understand the overall process and meaning of the Navier-Stokes equation. So, the objective of this report is to illustrate the equation. For an incompressible fluid (constant density, ρ), the full equation is expressed in Equation 1, where $\vec{u}$ means velocity in the control volume, p means pressure, μ means dynamic viscosity, and g means gravity (more generally, an external force).

$$\begin{aligned} m\vec{a} &= \vec{F'}_{\text{surface}} + \vec{F'}_{\text{body}} \\ \frac{m}{\Delta V}\vec{a} &= \frac{\vec{F'}_{\text{surface}}}{\Delta V} + \frac{\vec{F'}_{\text{body}}}{\Delta V} \implies \rho\vec{a} = \vec{F}_{\text{surface}} + \vec{F}_{\text{body}} \\ \rho\left(\frac{\partial\vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u}\right) &= -\nabla p + \mu\nabla^2\vec{u} + \rho\vec{g} \end{aligned} \tag{1}$$

2 Navier-Stokes Equation

2.1 Acceleration terms

For acceleration perspective, it is expressed as a time-dependent term (original acceleration) and a spatially dependent term (convection) in Equation 2.

$$\vec{a} = \frac{D\vec{u}}{Dt} = \frac{\partial\vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u} \tag{2}$$

2.2 Force terms

2.2.1 Body force

Body force means force from an external field (gravity, external force, and so on) regardless of interaction between particles in a fluid in Equation 3.

$$\vec{F}_{body} = \rho\vec{g} + f_{external}\cdots \tag{3}$$

2.2.2 Surface force

Surface force means force from interaction between particles, expressed with stress tensor in Equation 4. In stress tensor, the first index means normal vector for certain surface that a force is applied. The second index means force direciton at that surface. The stress tensor is consisting of pressure term and viscous stress tensor.

$$\begin{aligned} \vec{F}_{surface} &= \nabla\cdot\boldsymbol{\sigma},\ \boldsymbol{\sigma}\in\mathbb{R}^{3\times3} \\ \boldsymbol{\sigma} &= \begin{bmatrix}\sigma_{xx} & \sigma_{xy} & \sigma_{xz}\\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz}\\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz}\end{bmatrix} \\ \boldsymbol{\sigma} &= -p\mathbf{I} + \boldsymbol{\tau},\ \boldsymbol{\tau}\in\mathbb{R}^{3\times3} \end{aligned} \tag{4}$$

For an incompressible fluid, the viscous stress tensor is expressed in Equation 5, where λ denotes the second coefficient of viscosity (since density is constant).

$$\begin{aligned} \tau_{ij} &= \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right),\quad \tau_{xx} = 2\mu\frac{\partial u_x}{\partial x},\ \tau_{xy} = \mu\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) \\ \boldsymbol{\tau} &= \mu\left(\nabla\vec{u} + (\nabla\vec{u})^T\right) + \lambda(\nabla\cdot\vec{u})\mathbf{I} = \mu\left(\nabla\vec{u} + (\nabla\vec{u})^T\right),\ \because\nabla\cdot\vec{u} = 0 \end{aligned} \tag{5}$$

Therefore, divergence of stress tensor is expressed in Equation 6. Finally, Navier-Stokes equation for constant density is expressed in Equation 1.

$$\nabla \cdot \boldsymbol{\sigma} = -\nabla p + \mu \nabla^2 \vec{u} \tag{6}$$

2.3 Objective

The objective of this equation is to find velocity vector in control volume based on initial and boundary condition. For pressure, under the specific velocity, pressure is determined. Get divergence of the equaiton is expressed in Equation 7.

$$\begin{aligned}
&\nabla \cdot \left[\rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla)\vec{u} \right) \right] = \rho \frac{\partial}{\partial t} (\underbrace{\nabla \cdot \vec{u}}_{=0}) + \rho \nabla \cdot [(\vec{u} \cdot \nabla)\vec{u}] = \rho \nabla \cdot [(\vec{u} \cdot \nabla)\vec{u}] \\
&\nabla \cdot \left(-\nabla p + \mu \nabla^2 \vec{u} + \rho \vec{g} \right) = \nabla \cdot (-\nabla p) + \mu \nabla^2 (\nabla \cdot \vec{u}) + \rho \nabla \cdot \vec{g} = -\nabla^2 p \\
&\therefore \nabla^2 p = -\rho \nabla \cdot [(\vec{u} \cdot \nabla)\vec{u}]
\end{aligned} \tag{7}$$

3 Numerical methods

To solve the continuous Navier-Stokes equations computationally, the continuous spatial domain must be discretized into a finite set of discrete locations, known as a **grid** or **mesh**. Depending on how the physical variables (e.g., velocity $\vec{u}$, pressure p) are assigned and evaluated within this mesh, numerical methods are categorized into three major approaches:

- **Grid Points (Nodes):** Values are evaluated at discrete points. Used primarily in the **Finite Difference Method (FDM)**.
- **Elements & Nodes:** The domain is divided into sub-regions (elements), and shape functions describe variations within each element. Used in the **Finite Element Method (FEM)**.
- **Control Volumes (Cells):** The domain is partitioned into contiguous cells, and conservation laws are enforced over fluxes crossing cell faces. Used in the **Finite Volume Method (FVM)**.

Below is a detailed breakdown of each method.

3.1 FDM (Finite Difference Method)

FDM approximate derivative to basic operations.

$$\begin{aligned}
&\frac{\partial u}{\partial x} \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x} \\
&\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2} \\
&\frac{\partial u}{\partial t} \approx \frac{u_i^{n+1} - u_i^n}{\Delta t}, \quad u_i^{n+1} = u_i^n + \Delta t \times F(u_i^n, u_{i+1}^n, u_{i-1}^n) \\
&F(u_i^n, u_{i+1}^n, u_{i-1}^n) = -u_i^n \left(\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} \right) + \frac{\mu}{\rho} \left(\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta x)^2} \right) - \frac{1}{\rho} \left(\frac{p_{i+1}^n - p_{i-1}^n}{2\Delta x} \right) + g
\end{aligned} \tag{8}$$

3.2 FEM (Finite Element Method)

FEM does not approximate the derivative directly. Instead, it approximates the solution itself with shape functions in each element. First, multiply the 1D momentum equation by a test function $w(x)$ and integrate over the domain Ω. The second derivative term is lowered by integration by parts, so it is called **weak form** in Equation 9.

$$\begin{aligned}
&\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\mu}{\rho}\frac{\partial^2 u}{\partial x^2} - \frac{1}{\rho}\frac{\partial p}{\partial x} + g \\
&\int_\Omega w\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x}\right) dx = -\frac{\mu}{\rho}\int_\Omega \frac{\partial w}{\partial x}\frac{\partial u}{\partial x}\, dx + \left[\frac{\mu}{\rho}\, w\frac{\partial u}{\partial x}\right]_{\partial\Omega} + \int_\Omega w\left(-\frac{1}{\rho}\frac{\partial p}{\partial x} + g\right) dx
\end{aligned} \tag{9}$$

The velocity is expressed as a linear combination of shape functions $N_j(x)$ and nodal values $u_j(t)$. For a linear element, N_j is a hat function ($N_j(x_i) = \delta_{ij}$). In Galerkin method, the test function is the same as the shape function ($w = N_i$). Then, the weak form becomes a matrix equation in Equation 10, where $\mathbf{M}$ means mass matrix, $\mathbf{C}$ means convection matrix, $\mathbf{K}$ means stiffness (diffusion) matrix, and $\mathbf{F}$ means force vector.

$$\begin{aligned}
&u(x,t) \approx \sum_j N_j(x)\, u_j(t), \quad w = N_i \\
&M_{ij} = \int_\Omega N_i N_j\, dx, \quad C_{ij}(\mathbf{u}) = \int_\Omega N_i \left(\sum_k N_k u_k\right)\frac{\partial N_j}{\partial x}\, dx, \quad K_{ij} = \frac{\mu}{\rho}\int_\Omega \frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x}\, dx \\
&F_i = \int_\Omega N_i\left(-\frac{1}{\rho}\frac{\partial p}{\partial x} + g\right) dx + \left[\frac{\mu}{\rho}\, N_i\frac{\partial u}{\partial x}\right]_{\partial\Omega} \\
&\mathbf{M}\frac{d\mathbf{u}}{dt} + \mathbf{C}(\mathbf{u})\mathbf{u} + \mathbf{K}\mathbf{u} = \mathbf{F}, \quad \mathbf{M}\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = \mathbf{F}^n - \mathbf{C}(\mathbf{u}^n)\mathbf{u}^n - \mathbf{K}\mathbf{u}^n
\end{aligned} \tag{10}$$

3.3 FVM (Finite Volume Method)

FVM integrates the equation over each cell (control volume) $[x_{i-1/2}, x_{i+1/2}]$. The unknown is the cell average value $\bar{u}_i$, not the point value. To use divergence theorem, the convection term is rewritten as conservative form ($u\,\partial u/\partial x = \partial(u^2/2)/\partial x$). Then, the volume integral of the derivative becomes the difference of flux at both cell faces in Equation 11.

$$\begin{aligned}
&\bar{u}_i = \frac{1}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} u\, dx, \quad \frac{\partial u}{\partial t} + \frac{\partial \mathcal{F}}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + g, \quad \mathcal{F} = \frac{u^2}{2} - \frac{\mu}{\rho}\frac{\partial u}{\partial x} \\
&\frac{d\bar{u}_i}{dt} = -\frac{\mathcal{F}_{i+1/2} - \mathcal{F}_{i-1/2}}{\Delta x} - \frac{1}{\rho}\frac{p_{i+1/2} - p_{i-1/2}}{\Delta x} + g
\end{aligned} \tag{11}$$

The face values are not known, so they are interpolated from neighboring cells in Equation 12. Since the flux leaving cell i is exactly the same as the flux entering cell $i+1$, the conservation is satisfied in discrete level.

$$\begin{aligned}
&u_{i+1/2} \approx \frac{\bar{u}_i + \bar{u}_{i+1}}{2}, \quad \left(\frac{\partial u}{\partial x}\right)_{i+1/2} \approx \frac{\bar{u}_{i+1} - \bar{u}_i}{\Delta x}, \quad p_{i+1/2} \approx \frac{p_i + p_{i+1}}{2} \\
&\mathcal{F}_{i+1/2} \approx \frac{1}{2}\left(u_{i+1/2}\right)^2 - \frac{\mu}{\rho}\frac{\bar{u}_{i+1} - \bar{u}_i}{\Delta x} \\
&\bar{u}_i^{n+1} = \bar{u}_i^n - \frac{\Delta t}{\Delta x}\left(\mathcal{F}_{i+1/2}^n - \mathcal{F}_{i-1/2}^n\right) - \frac{\Delta t}{\rho}\frac{p_{i+1/2}^n - p_{i-1/2}^n}{\Delta x} + \Delta t\, g
\end{aligned} \tag{12}$$

References

[1] Zhen Lei and Xiao Ren. Finite-time blowup for navier-stokes with smooth forcing, part i: Construction of self-similar solutions with admissible stress and flat remainder. *arXiv preprint arXiv:2609.35406*, 2026.